

Protasov's Lemma: Exercise Sheet

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Problem 1 (Thailand TST): Let ABC be a triangle and let D be a point on BC such that $B \in (CD)$. Let c_0 be the circumcircle of $\triangle ABC$ and c_1 the circle tangent to the segments (AD) , (DB) and to the circle c_0 . We denote by M, N the intersection of c_1 with DB and AD respectively; and by I_C - the excenter of the triangle $\triangle ABC$ (corresponding to C). Show that M, N, I_C are collinear.

Problem 2 (Romania TST 2002, Problem 2): Let ABC be a triangle with $AB < AC$, and c_0 its circumcircle. The tangent in A to c_0 intersects BC in a point D . Let c_1 be the circle tangent to the segments (AD) , (BD) and to the circle c_0 . We denote by M the intersection of c_1 and BD . Then:

$$AM \text{ is the bisector of } \angle(DAB) \iff CA = CM$$

Problem 3 (Archimedes' Lemma): Let C be a circle, and let C' be a circle internally tangent to C at point A . Suppose BC is a chord of C that is also tangent to C' at point D . Then the line AD bisects $\angle BAC$.

Problem 4: Let A, B be points on a circle S and C a point in its interior. Consider S' , tangent to the segments AC, BC and to the circle S in three points denoted P, Q and R respectively. Show that the circumcircle of $\triangle APR$ contains the incenter of $\triangle ABC$.

Problem 5: Let ABC a triangle with circumcircle $\odot(O, R)$. We consider ω the circle that is externally tangent to $\odot(O, R)$ and tangent to the half-lines $(AB), (AC)$, in the points F, K and L , respectively. Let D be the midpoint of the arc BC (containing A). We denote by E the intersection of DF and KL . Prove that E is the excenter of $\triangle ABC$ associated to A .

Problem 6 (All-Russian Olympiad 2010 11-th Grade Problem 3): Let $ABCD$ be a cyclic quadrilateral with circumcircle S , and $\{K\} = AC \cap BD$. Denote by M_1, M_2, M_3, M_4 the midpoints of the arcs AB, BC, CD and DA respectively (where each arc contains no other vertex of the quadrilateral). Let S_1 be the circle tangent to the segments KC, KD and to the arc CD . We consider analogously the circles S_2, S_3, S_4 corresponding to the arcs DA, AB and BC respectively. We denote $N_i = S_i \cap S$ for $i = \overline{1, 4}$. Prove that M_1N_1, M_2N_2, M_3N_3 and M_4N_4 are concurrent.

Problem 7: Let ABC be a triangle and $\odot(I, r)$ its incircle, tangent to BC in D . We consider w the circle passing through B, C and tangent to $\odot(I, r)$ in a point L . Prove that A is on the radical axis of the circles w and $\odot(IDL)$.

Problem 8: Let $ABCD$ be a cyclic quadrilateral with circumcircle C and let E be the intersection of its diagonals. The half-lines (CB) and (DA) intersect in a point F . Let I_1, I_2, I_3, I_4 be the incenters of the triangles ABE, ABF, CDE and CDF . Prove that I_1I_2 and I_3I_4 intersect on the circle C .

Problem 9: Let ABC be a triangle, H - its orthocenter, M - the midpoint of BC , K - the projection of H on AM and $\odot(I, r)$ - the incircle of $\triangle ABC$. The circles (AKI) and (BHC) intersect in K and another point X . Prove that the midpoint of AX is on the circle $\odot(I, r)$.

Problem 10: Let $ABCD$ be a cyclic quadrilateral with circumcircle C and let P be the intersection of its diagonals.

- Let ω_1 be the circle tangent to the half-lines $(PC), (PD)$ and internally tangent to C in the point X .
- Similarly, let ω_2 be the circle tangent to the half-lines $(PC), (PD)$ and externally tangent to C in Y .

Let Z the intersection point of PX and the incircle of $\triangle PCD$ (the point closer to X) and T the intersection between the segment PY and the excircle of $\triangle PCD$ corresponding to P . Let E, F be the contact points of the incircle of $\triangle PCD$ and the excircle of $\triangle PCD$ with BC .

Prove that Z, T, E, F lie on a circle centered in the midpoint of the arc CAD .

Problem 11: Let ABC be a triangle and $\odot(O, R)$ its circumcircle. Let ω be a circle tangent to $\odot(O, R)$ at A and tangent to BC . The other tangents to ω through B and C intersect in a point D . The line DA intersects $\odot(O, R)$ again in a point E . We take the tangents from E to ω and they intersect BC in G and H . Prove that $\odot(EGH)$ and $\odot(O, R)$ are tangent.

Problem 12: Let $\triangle ABC$ be a triangle, and let ω denote its circumcircle. Let D' be the point of tangency between side BC and the A -excrcle of triangle ABC . Let X be the point of tangency between ω and the A -mixtilinear incircle (i.e. the circle tangent to ω and to the segments AB and AC).

Prove that

$$\angle BAX = \angle CAD'.$$

Problem 13: Let $\triangle ABC$ be a triangle, and let ω be its circumcircle. Let D be the point of tangency between side BC and the incircle of triangle ABC . Let X be the point of tangency between ω and the A -mixtilinear circle (i.e., the circle tangent to ω and to the segments AB and AC).

Prove that

$$\angle BXA = \angle CXD.$$

Problem 14: Let $ABCDEF$ be a bicentric hexagon with incenter I and circumcenter O . The diagonals AD, BE and CF meet in a point (c.f. Brianchon's Theorem) denoted G . Let J, K, L, M, N, P be the incenters of the triangles GAB, GBC, GCD, GDE, GEF and GFA respectively. We consider the circle tangent to the arc AB and to the segments $(AG), (BG)$; and denote by X the tangency point with the arc AB . Analogously we define Y, Z, T, U, V on the arcs BC, CD, DE, EF and FA . Prove that the lines XM, YN, ZP, TJ, UK, VL meet in a point S on the line OI .

Problem 15: Let ABC be an acute triangle with circumcenter O and incenter I . Denote by T_A the tangency point of the A -mixtilinear circle with $\odot(ABC)$ and define T_B similarly. Let AO meet BC at D , BO meet AC at E . Let X be the intersection of lines AI and T_AD , Y be the intersection of lines BI and T_BE . Prove that CI is the angle bisector of $\angle XCY$.

Note: The A -mixtilinear circle is the circle tangent to the segments $(AB), (AC)$ and internally tangent to the circumcircle $\odot(ABC)$.

Problem 16: Let ABC be an acute triangle with I its incenter and ω its circumcircle. Consider the circle ω_B , which is tangent to sides BA and BC , and internally tangent to ω at point B_1 . The circle ω_C and point C_1 are defined analogously.

Let B_2 and C_2 be the points diametrically opposite to B and C on ω , and let $X = B_1C_2 \cap B_2C_1$. **Prove that $XA = XI$.**

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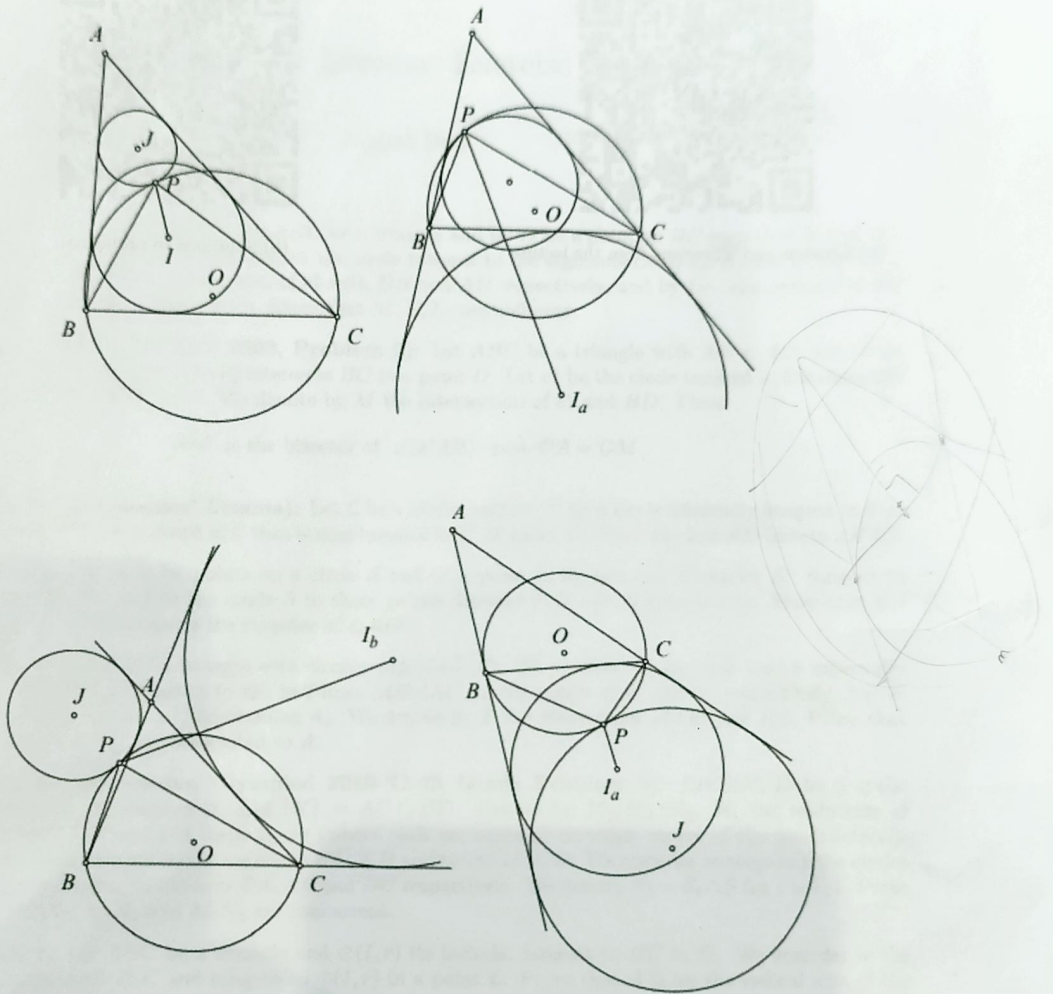


Figure 1: All configurations of Protassov's Lemma

This picture is taken from the article on Protassov's Lemma:

- Bài toán Protassov và ứng dụng by Nguyễn Văn Linh.

Using any Artificial Intelligence, you can make the contents of this article available in any language you desire.



(a) Lemmas and Theorems from the lecture



(b) Solutions to problems